An Internship Report

Of

Google AI-ML Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)

By

Roll Number of Student: 2410030981

Name of Student: PIYUSH KUMAR YADAV

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, **PIYUSH KUMAR YADAV**, do hereby declare that the internship report titled “Google AI-ML Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-
Student Name: PIYUSH KUMAR YADAV
Roll No.: 2410030981

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – EduSkills Virtual Internship Program team, EduSkills Academy, and Google for Developers for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Google AI-ML. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-
Student Name: PIYUSH KUMAR YADAV
Roll No.: 2410030981

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3. INTERNSHIP COMPLETION CERTIFICATE

शिक्षा मंत्रालय
MINISTRY OF
EDUCATION

NATIONAL
INTERNSHIP
PORTAL

EduSkills®
Nation Building Through Skills

AICTE - EduSkills
2026
VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Piyush Yadav

IILM University, Greater Noida

has successfully completed the 8-weeks

AI-ML Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : **India Edu Program**
Google for Developers

Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 479384cd2c13d79088f5
Student ID: STU6997eaa57f1f91771563685

INTERNSHIP GRADE
E

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter

EduSkills®
Nation Building Through Skills

Ref No: C16Y2026M081377131

<u>Internship Offer Letter</u>

Student Name	: Piyush Yadav
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Google AI-ML
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU6997eaa57f1f91771563685

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “Google AI-ML”, supported by Google for Developers. The internship was carried out from April 2026 to June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong practical foundation in Machine Learning and Deep Learning using TensorFlow, and to apply it to real-world computer-vision problems such as image classification, on-device object detection, and product image search for mobile applications.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on assignments, quizzes, learning-pathway badges, and a final credential validation. On successful completion of all the requirements, including the Internship Grade Point Assessment, a Virtual Internship Certificate with an “Excellent” grade was awarded by EduSkills Academy under the patronage of AICTE and the National Internship Portal.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Google AI-ML track, EduSkills collaborates with Google for Developers to design an industry-aligned curriculum, delivered through the Google Developer Program’s learning pathways, quizzes, and skill badges, that helps engineering students build practical machine-learning and computer-vision capability.

4.3 Problem Statement

Although a large number of engineering students learn the theoretical foundations of machine learning and neural networks during their academic coursework, very few get structured, hands-on exposure to building and applying computer-vision models using an industry-standard framework such as TensorFlow. Concepts such as convolutional neural networks, on-device object detection, and image classification are often understood only at a theoretical level, with little practice in training, converting, and integrating such models into real applications. The problem addressed by this internship was to practically learn and apply the TensorFlow-based machine-learning workflow – from building and training a neural network, to detecting and classifying objects on-device, to building a product image search capability for a mobile application.

4.4 Project Objectives

- To understand the first principles of machine learning and build convolutional neural networks (CNNs) using TensorFlow for image recognition and classification.
- To learn the basics of object detection and integrate pretrained object detectors into mobile applications.

- To train custom object-detection models using TensorFlow Lite and the TFLite Model Maker.
- To build a product image search feature using on-device object detection.
- To detect objects in static images and live camera feeds for visual product search.
- To build backend integration for a product image search mobile application.
- To build custom image-classification models and improve overall classification skill.
- To create and integrate a custom image-classifier model into an application.To successfully clear the Final Credential Validation / Internship Grade Point Assessment.

4.5 Scope of the Project

The scope of this internship spans the complete TensorFlow-based computer-vision workflow – starting from understanding neural-network fundamentals and building CNNs, moving to on-device object detection using pretrained and custom-trained models, and extending into a product image search feature that combines object detection with visual search across static images and live camera feeds. It further covers building and improving custom image-classification models and integrating them into an application. The internship was limited to a virtual, simulation/lab-based learning environment provided by the Google Developer Program and EduSkills Academy, and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Programming & ML Framework: Python, TensorFlow, Keras
- On-device Machine Learning: TensorFlow Lite, TensorFlow Lite Model Maker
- Computer Vision: Convolutional Neural Networks (CNNs), pretrained object detectors, custom object-detection models, custom image classifiers
- Application Layer: Mobile application integration, camera-feed based live object detection
- Learning Platform: Google Developer Program learning pathways, quizzes, and skill badges

4.7 System Architecture

The general architecture followed during the internship for the Google AI-ML learning track can be summarized in the following layers:

- Model Layer: A convolutional neural network built and trained using TensorFlow forms the core intelligence layer for image recognition and classification.
- Conversion Layer: Trained models are converted into a lightweight, on-device format using TensorFlow Lite and the TFLite Model Maker.
- Detection Layer: Pretrained and custom-trained object detectors identify and localize objects within static images and live camera feeds.
- Search Layer: A product image search feature combines on-device object detection with backend integration to enable visual product search.
- Classification Layer: A custom image classifier further categorizes detected objects/images with improved accuracy.

- Application Layer: The trained and converted models are integrated into a mobile application, consuming the detection, search, and classification capabilities.
- Assessment Layer: Weekly quizzes, learning-pathway badges, and a Final Credential Validation / Internship Grade Point Assessment track progress and final grading.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and quiz-based assessments, culminating in a Final Credential Validation, as summarized in the table below.

Week	Module	Description
week 1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from first principles to creating convolutional neural networks for advanced image recognition and classification.

Week	Module	Description
week 2	Get started with object detection	Basics of object detection and integrating pretrained object detectors into mobile apps.
week 3	Go further with object detection	Train custom object-detection models using TensorFlow Lite and Model Maker.
week 4	Get started with product image search	Build a product image search feature using on-device object detection.
week 5	Get started with product image search	Detect objects in static images and live camera feeds for visual product search.
week 6	Go further with product image search	Build backend integration for product image search mobile applications.
week 7	Go further with image classification	Build custom image-classification models and improve classification skills.
week 8	Go further with image classification	Create and integrate a custom image classifier model into your app.
Final Credential Validation / Internship Grade Point Assessment		

Note: There was no stipend associated with this internship.

Each week concluded with quiz-based learning-pathway assessments and skill badges to evaluate conceptual understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

4.9 Expected Outcomes

- Ability to explain and apply the first principles of machine learning to build convolutional neural networks using TensorFlow.
- Practical understanding of object detection, including the use of pretrained detectors and custom-trained TensorFlow Lite models.

- Hands-on exposure to building a product image search feature combining on-device detection with backend integration.
- Capability to build, evaluate, and improve custom image-classification models.
- Skill to integrate trained and converted TensorFlow Lite models into a mobile application.
- A completed set of weekly learning-pathway modules and skill badges, along with successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Excellent” grade.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Google AI-ML, are attached in Chapter 3 of this report. The official completion e-mail from EduSkills Foundation and the corresponding skill badges earned on the Google Developer Program profile are attached below as further communication proof.

AICTE-EduSkills Internship Certificate download link

External | Inbox

EduSkills Foundation <noreply@eduskillsfoundation.org>
to me

Wed, Jul 8, 12:05 AM

Dear **Piyush Yadav**,

🎉 **Congratulations!**

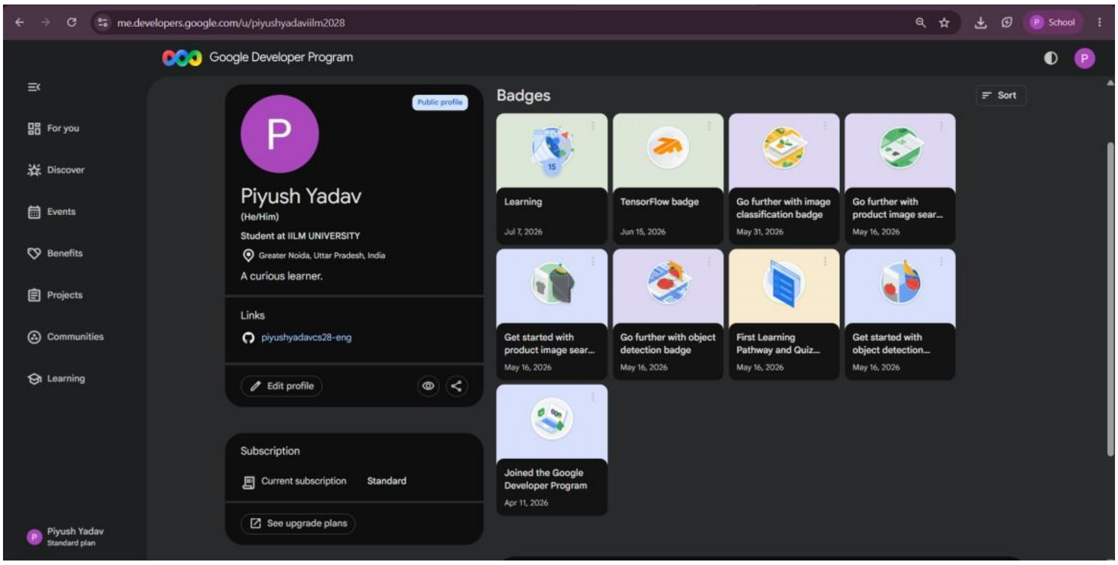
You have successfully completed **10 weeks Google AI-ML** internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.

📥 **Steps to Download Your Certificate:**

1. Login at: https://eduskillsfoundation.org/login
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

Reply | Forward | New | Share in chat



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